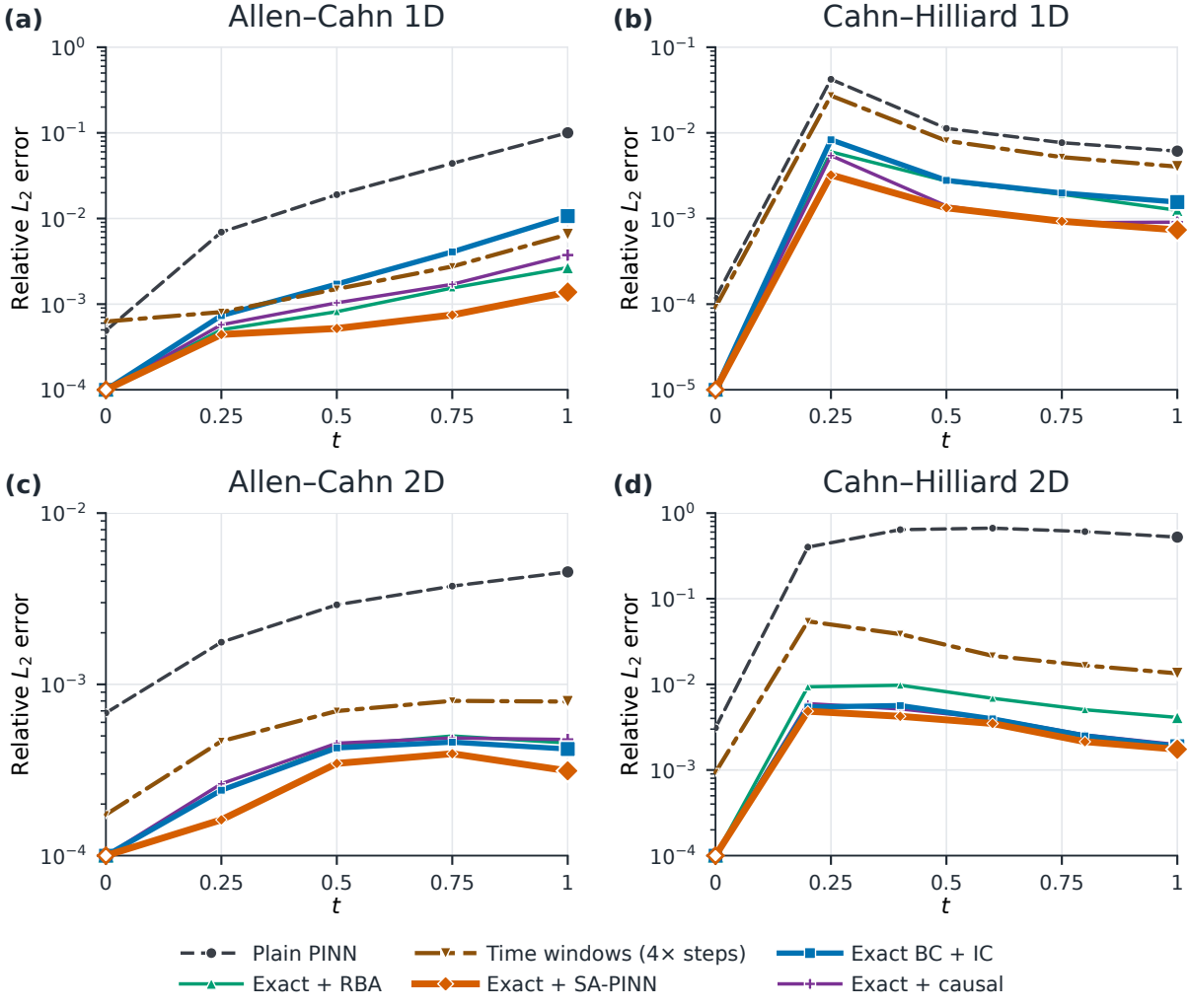
(a)
Allen–Cahn 1D
(b)
Cahn–Hilliard 1D
(c)
Allen–Cahn 2D
(d)
Cahn–Hilliard 2D
Relative L_2 error
t
Plain PINN
Time windows (4× steps)
Exact BC + IC
Exact + RBA
Exact + SA-PINN
Exact + causal